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BeeHive- Purple Bee final Documentation

Thursday 10 September 2026

Appsolutely Software



Requirements Document



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Project Overview

System overview and objective

BeeHive is X Spark's internal meeting application. It supports immediate Start Now meetings and scheduled meetings, invitations, participants, agendas, recordings, transcripts, personal AI-generated notes, and suggested action items; Purple Bee is X Spark's internal work-management application for teams, projects, tasks, assignments, notifications, and task activity.

Problem and Selected Solution: Today, a decision made in BeeHive must be entered again in Purple Bee. This causes duplicate capture, inconsistent wording, missing ownership, and weak traceability between a meeting and the resulting work. The integrated solution will allow confirmed BeeHive action items to become PurpleBee tasks through authenticated server-to-server HTTP APIs. The HTTP interface will always run over HTTPS in development, staging, and production.

System boundary: BeeHive remains the source of meeting context, agenda sources, permitted AI-note sources, the human confirmation record, and transfer status. Purple Bee remains the source of truth for teams, projects, profiles, created tasks, assignment notifications, and the later task lifecycle. The systems exchange immutable Universally Unique Identifiers (UUIDs) through APIs; they do not use cross-database foreign keys or write directly into one another's databases.

Confirmed trigger: Meeting completion and AI-processing completion prepare reviewable candidates. They do not create Purple Bee tasks. The actual business trigger is an explicit confirmation of one action item by the nominated project lead whose active Purple Bee team role is Admin or Manager.

Business question	Completed answer
What Purple Bee does today	Manage team projects, tasks, assignees, task state, notifications, and activity history.
What BeeHive does today	Creates and runs meetings; stores participants, agendas, recordings, transcripts, personal AI notes, and suggested actions.
Why integrate	Remove duplicate task capture and preserve traceability from confirmed meeting action to operational task.
Cross-boundary process	Review a completed meeting, select or edit a candidate, map project and assignee, confirm, deliver by HTTP API, and return

	the task UUID
Information that moves	Approved task title and description, priority, due date, assignee UUID, project UUID, tags, meeting UUID, canonical action UUID, confirmer identity, and confirmation timestamp.
New user capability	A nominated admin or manager confirms once in BeeHive and receives a linked Purple Bee task without retyping it.
Explicit first-phase exclusions	Automatic task creation without confirmation, two-way synchronisation, direct database access, transfer of recordings, transcripts or full AI summaries, private notes, and automatic replacement of missing or deleted tasks.

Business value and measurable outcomes

Measure	Required outcome
Duplicate prevention	One confirmed action item creates no more than one Purple Bee task under retries, replay, or concurrent confirmation.
Traceability	BeeHive stores the source meeting and action UUIDs, confirmer audit, and returned Purple Bee task UUID.
Control	Only the nominated project lead with an active Admin or Manager team role can confirm.
Privacy	No recording, full transcript, full AI summary, secret, or unrelated private note is transferred.
Efficiency	Pilot measurement compares manual capture time with confirmed-action transfer time; the baseline is captured before UAT.

System assumptions

Assumption	Status	Evidence or validation
Both applications continue to use separate Supabase-backend environments.	Confirmed	Repositories and selected architecture
The confirmer has a BeeHive identity mapped to one Purple Bee profile UUID.	To Validate	Identity mapping design

The nominated confirmer is an active Admin or Manager of Purple Bee.	Confirmed rule	Stakeholder decision
Saved agendas are shared meeting data; personal_meeting_notes remain participant-private.	Confirmed	BeeHive repository
Representative non-sensitive staging data and separate credentials will be available.	To Validate	Security and test readiness
The first release is one-way task creation; later Purple Bee edits are not written back.	Confirmed	Selected Scope
Purple Bee derives team_id from project_id and ignores or rejects conflicting client data.	Confirmed design	Target ownership rule

Stakeholders and actors

Stakeholder / Actor	Role	Responsibilities	System Interaction
Business Owner/Product Owner	Decision maker	Defines business goals, scope and priorities	Reviews requirements, scope, priorities and business outcomes.
BeeHive User	Meeting participant	Creates meetings, agendas and action items	Uses BeeHive to manage meetings and review action items.
Project Lead (Admin/Manager)	Authorised confirmer.	Reviews and confirms meeting action items for transfer to Purple Bee.	Confirms approved action items in BeeHive for transfer.
Purple Bee User	Task assignee	Completes assigned tasks and updates task progress.	Uses Purple Bee to view and manage assigned tasks.
Purple Bee Manager	Team manager	Oversees projects, tasks, team members and approvals.	Uses Purple Bee to manage projects, tasks and team activities.
Purple Bee Administrator	System administrator	Manages users, permissions and system configuration	Uses administrative functions to manage system access and configuration.
Integration Service	System actor	Transfers approved action items between systems	Executes authenticated API requests, transfers approved action items, and records transfer results.

Development Team	Technical stakeholder	Develops, tests, documents and maintains the integration.	Develops and supports the integration across development, staging and production environments.
Supabase Services	External service	Provides authentication, database and realtime services required by the applications.	Supports backend authentication, data storage and realtime functionality.

Development Team

Team Member	Project Role	Responsibilities
Thimna Booi	Project Lead / Documentation	Coordinates the project, tracks progress and manages project documentation.
Kelly Nolte	Researcher / Integration Developer	Conducts backend research and development, designs the API integration, implements security and supports testing.
Kgothatso Kaole	Database Developer / Data Analyst	Analyses data requirements, supports database design and ensures the required data is correctly mapped between systems.

Devices and System Overview

Component	Description
User Device	Users access BeeHive and Purple Bee through supported web-enabled devices.
BeeHive	Meeting and collaboration application that manages meetings, participants, agendas and action items.
Integration Service	Backend service responsible for securely transferring confirmed action items.
Purple Bee	Work-management application responsible for projects, tasks, assignments and notifications.
Database Services	Separate backend databases used by the applications.

API	Authenticated HTTPS API used for communication between BeeHive and Purple Bee.

Functional Requirements

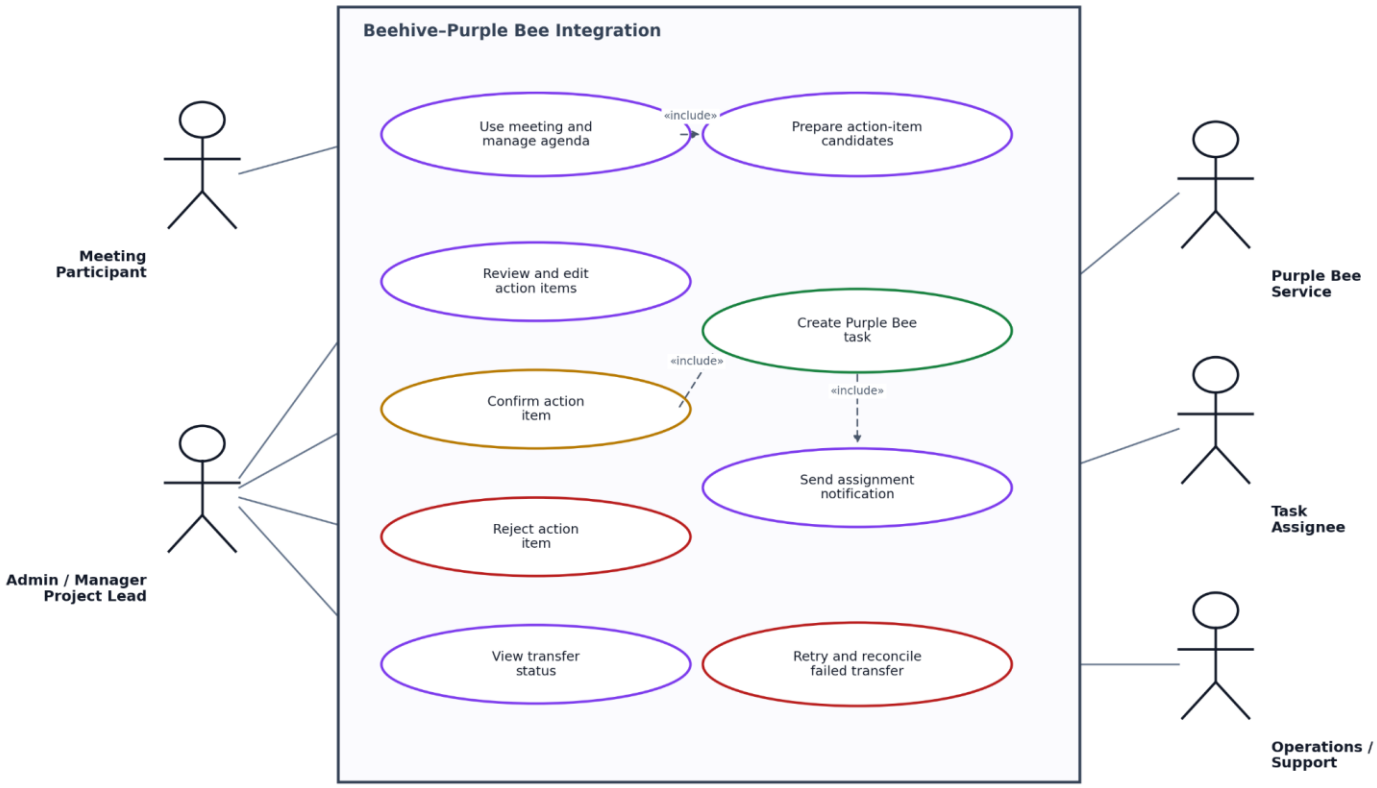
ID	Business rule	Source or evidence	Priority
BR-001	Meeting completion prepares candidates and never creates a Purple Bee task.	BeeHive repository and selected option	High
BR-002	AI completion prepares permitted candidates and never automatically creates a task.	Selected option	High
BR-003	Only the nominated project lead with active Purple Bee Admin or Manager team authority may confirm.	Stakeholder decision	High
BR-004	Purple Bee re-checks the confirmer's role, membership, project, assignee, and required values on every create request.	Security design	High
BR-005	Every exportable item has a stable BeeHive meeting_action_items_id UUID.	Integration design	High
BR-006	Agenda items[].completed describes agenda coverage and is not task confirmation.	BeeHive repository	Medium
BR-007	Only the confirmer's own private AI-note source may be used unless a separate sharing feature is approved.	BeeHive repository and privacy rule	High
BR-008	Title, project, assignee, priority, and non-past due date are required before confirmation.	V1 contract	High
BR-009	Purple Bee derives team_id from the selected project.	Target ownership	High
BR-010	The idempotency key must follow the format beehive:{meeting_id}:{action_item_id}.	Reliability design	High
BR-011	The same key and request hash returns the existing task; a changed request with the same key returns 409.	Reliability design	High
BR-012	Task, integration receipt, and required assignment notification commit atomically.	Target transaction design	High

BR-013	A new meeting-derived task starts as todo, progress 0, and non-recurring.	v1 mapping	Medium
BR-014	Recordings, full transcripts, full summaries, secrets, and unrelated participant data never cross the API.	Security and privacy standard	High
BR-015	Purple Bee owns the created task lifecycle; BeeHive owns meeting provenance and transfer state.	Architecture design	High
BR-016	Permanent validation errors return the item for correction; only approved transient errors are retried.	Reliability design	High
BR-017	Reconciliation never silently creates a replacement for a missing or deliberately deleted task.	Operations design	Medium
BR-018	All HTTP API traffic uses HTTPS and environment-specific authentication credentials.	Confirmed HTTP design and security standard	High

Business use case model

The model covers normal BeeHive meeting use, action-item and confirmation, HTTP task creation, transfer status, task notification, and operational recovery. The boundary separates user action from the integration worker and external services.

Business Use Case Model — Beehive and Purple Bee Integration



One-way task creation: the Admin/Manager confirmation triggers Beehive to send the approved action item to Purple Bee.

Capability group	Included capability
Authentication and access	Authenticate BeeHive users; validate nomination and Purple Bee Admin or Manager authority.
Core BeeHive functions	Create or schedule meetings, manage agenda, complete meeting, process permitted AI notes, and review candidates.
Core Purple Bee functions	Return project/member context, create task, record receipt, and notify assignee.
Cross-system workflow	Confirm one action, queue delivery, call HTTP API, and store returned task UUID.
Data integration	Map immutable UUIDs, normalise fields, prevent duplicates, and reconcile receipts.
Administration and reporting	Manage mappings, feature flag, credentials, metrics, failures, and audit evidence.
Exception handling	Correct validation failures; retry transient failures, dead-letter, and investigate unresolved records.

User activities

The following user activities describe the main interactions involved in the BeeHive-to-Purple Bee integration. These activities provide a foundation for the Use Case Model, Use Case Glossary, Use Case Narratives, and Activity Diagrams.

UA-01: Review Meeting Action Items

Actor: BeeHive User / Project Lead

Activity:
After a meeting has been completed and permitted AI processing has finished, the user reviews the suggested action items in BeeHive. The system presents the action items as candidates for confirmation and does not automatically create Purple Bee tasks.

UA-02: Select and Edit an Action Item

Actor: Project Lead (Admin/Manager)

Activity:
The nominated project lead selects an action item for transfer and reviews or edits the task

information. The project, assignee, priority, due date, title, and description are checked before confirmation.

UA-03: Confirm an Action Item

Actor: Project Lead (Admin/Manager)

Activity:
The nominated project lead explicitly confirms an action item for transfer. BeeHive records the confirmer identity and confirmation timestamp and prepares the approved action item for delivery to Purple Bee.

UA-04: Transfer Action Item to Purple Bee

Actor: Integration Service

Activity:
The integration service sends the confirmed action item to Purple Bee using an authenticated HTTPS API request. The request contains the approved task information and the relevant BeeHive and Purple Bee UUIDs.

UA-05: Validate and Create Purple Bee Task

Actor: Purple Bee / Integration Service

Activity:
Purple Bee validates the request, including the confirmer's authority, project, assignee, required values, and idempotency information. If validation succeeds, Purple Bee creates the task and returns the task UUID.

UA-06: Record Transfer Result

Actor: Integration Service / BeeHive

Activity:
The integration records the result of the transfer, including the BeeHive action-item UUID, meeting UUID, confirmation information, transfer status, and returned Purple Bee task UUID.

UA-07: Notify the Assignee

Actor: Purple Bee

Activity:
After successful task creation, Purple Bee creates the required task-assignment notification for the selected assignee.

UA-08: Handle Transfer Failure

Actor: Integration Service / Project Lead

Activity:
If the transfer fails, the system records the failure. Permanent validation errors return the action item for correction, while approved transient errors may be retried. Unresolved failures are recorded for operational investigation.

Use Case Glossary & Narratives

Use case glossary

Use Case ID	Name	Actor	Purpose
UC-001	Link meeting to Purple Bee project	Admin or Manager project lead	Choose valid project context and nominate confirmer.
UC-002	Prepare action item candidates	BeeHive backend	Materialise reviewable agenda, permitted AI, and manual candidates.
UC-003	Review and confirm action item	Admin or Manager project lead	Validate and approve one item for export.
UC-004	Create Purple Bee task through HTTP API	Integration worker	Create exactly one target task and return its UUID.
UC-005	View transfer result	Authorised meeting user	See queued, created, correction, or failure state.
UC-006	Retry failed delivery	Integration worker or operator	Retry only eligible transient failures.
UC-007	Reconcile integration records	Operations or Support	Identify missing or inconsistent source and target records.
UC-008	Monitor and administer integration	Operations or Support	Use metrics, alerts, logs, feature flag, and credential controls.

UC-009	Reject action item	Admin or Manager project lead	Record human rejection without exporting.
UC-010	Receive task assignment notification	Task assignee	Receive the existing Purple Bee task notification.

Integration-specific narratives

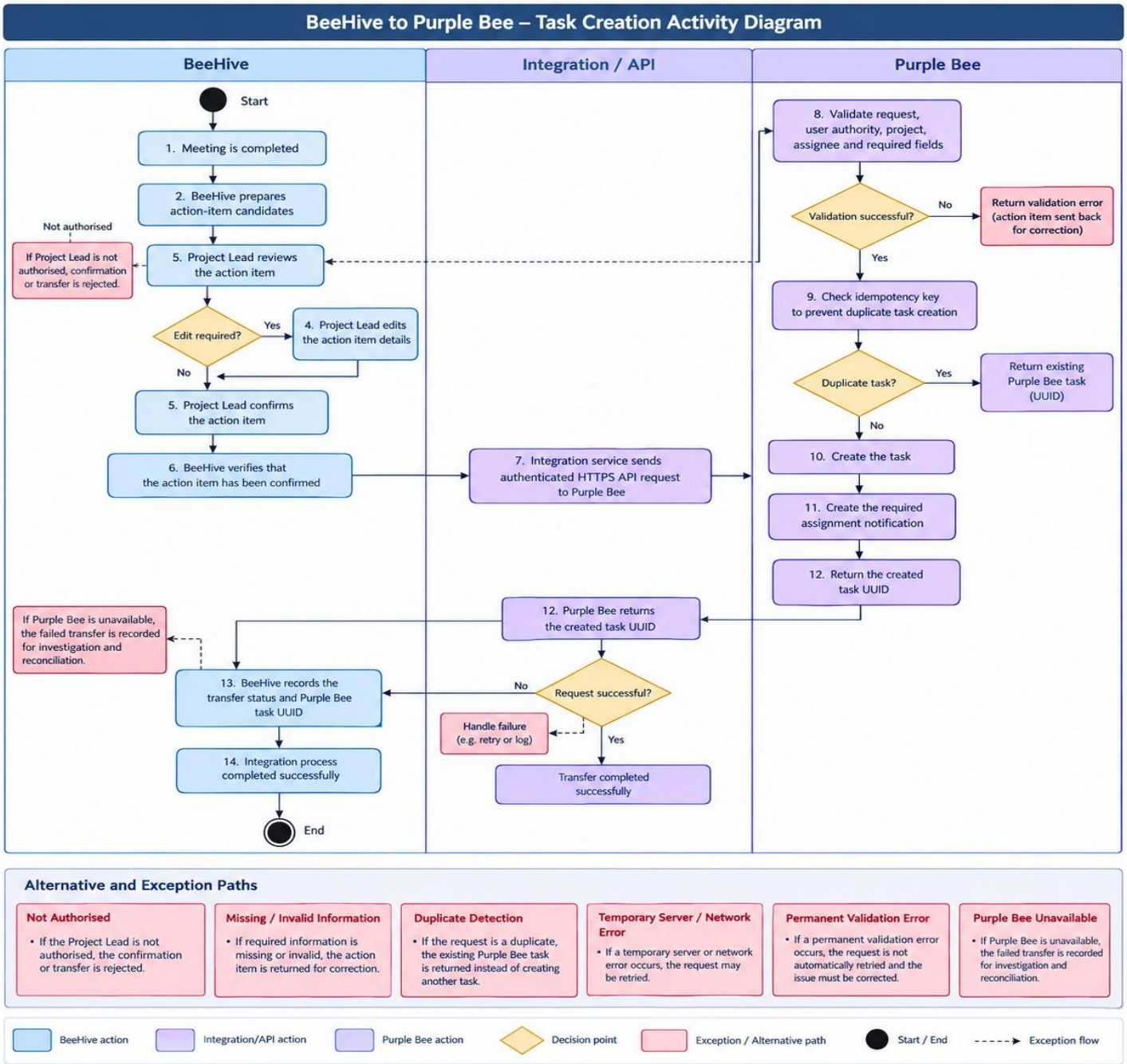
UC-001 Link meeting to Purple Bee project

Field	Completed specification
Use Case ID and name	UC-001 Link meeting to Purple Bee project
Primary actor	Nominated Admin or Manager project lead
Supporting actors	Beehive backend, Purple Bee beehive-context-v1 HTTP API and Supabase Auth
Goal	Create an active, validated link between one Beehive room and a Purple Bee project, team and confirmer profile.
Preconditions	Actor can access the meeting and has a mapped Purple Bee profile.
Trigger	Actor opens integration setup or an existing mapping requires revalidation.
Main success flow	1. Beehive authenticates the actor. 2. Beehive calls the HTTPS context API. 3. Purple Bee returns projects and active members available to the mapped profile. 4. Actor selects project and confirmer. 5. Beehive stores external UUIDs and validation audit.
Alternative or exception flows	401 invalid service authentication; 403 mapped profile lacks access; empty project list; stale project or member is rejected and the mapping remains inactive.
Business rules	BR-003, BR-004, BR-009 and BR-018.
Data created or changed	meeting_project_links with room_id, purplebee_team_id, purplebee_project_id, both confirmer IDs, status and verification fields.
Postconditions	An active mapping exists and may be used to prepare and confirm actions.
Acceptance criteria	Given an authorised mapped user, the API returns only permitted project/member UUIDs and the selected mapping is saved. A forbidden user receives 403 and no active mapping is created.

UC-003 Review and confirm action item

Field	Completed specification
Use Case ID and name	UC-003 Review and confirm action item
Primary actor	Nominated project lead with active Purple Bee Admin or Manager team role
Supporting actors	Beehive backend, context API, and Supabase Auth
Goal	Approve accurate, complete, and permitted task information for export.
Preconditions	Meeting and canonical action exist; active project link exists; actor can access the meeting.
Trigger	Actor selects Confirm on one reviewed item.
Main success flow	1. Beehive authenticates the session. 2. It loads the action and active project link. 3. It confirms the actor is the nominated confirmer. 4. It validates title, project, assignee, priority, due date, source permission and version. 5. One transaction marks the action confirmed and inserts one outbox event. 6. Beehive returns 202 Accepted.
Alternative or exception flows	Invalid field returns correction details; stale version returns 409; unauthorised actor returns 403; rejection records rejected status and creates no outbox event.
Business rules	BR-001 to BR-010, BR-014 and BR-018.
Data created or changed	meeting_action_items review status, confirmed_by, confirmed_at, version, idempotency_key and transfer status; integration_outbox row.
Postconditions	Exactly one create intent is queued; a Purple Bee task is not yet claimed.
Acceptance criteria	A valid nominated Admin or Manager receives 202, and one outbox row exists. A regular or non-nominated user receives 403, and no outbox row is created.

Activity Diagram



BeeHive to Purple Bee Task Creation Activity

The activity diagram represents the end-to-end process of transferring a confirmed BeeHive action item to Purple Bee. The process begins when a meeting is completed and action-item candidates are available for review. The nominated Project Lead reviews and confirms an action item before it can be transferred.

The process follows these steps:

1. Meeting is completed.
2. BeeHive prepares action-item candidates.
3. Project Lead reviews the action item.
4. Project Lead edits the action item details if required.
5. Project Lead confirms the action item.
6. BeeHive verifies that the action item has been confirmed.
7. The integration service sends an authenticated HTTPS API request to Purple Bee.
8. Purple Bee validates the request, user authority, project, assignee and required fields.
9. Purple Bee checks the idempotency key to prevent duplicate task creation.
10. If validation is successful, Purple Bee creates the task.
11. Purple Bee creates the required assignment notification.
12. Purple Bee returns the created task UUID.
13. BeeHive records the transfer status and Purple Bee task UUID.
14. The integration process is completed successfully.

Alternative and Exception Paths

- If the Project Lead is not authorised, the confirmation or transfer is rejected.
- If required information is missing or invalid, the action item is returned for correction.
- If the request is a duplicate, the existing Purple Bee task is returned instead of creating another task.
- If a temporary server or network error occurs, the request may be retried.
- If a permanent validation error occurs, the request is not automatically retried and the issue must be corrected.
- If Purple Bee is unavailable, the failed transfer is recorded for investigation and reconciliation.

Diagram requirement: The activity diagram should show the three main system boundaries: **BeeHive** → **Integration/API** → **Purple Bee**, including decision points for authorisation, validation, duplicate detection and failure handling.

Interface Requirements

API Interface

The BeeHive and Purple Bee systems will communicate through an authenticated REST API. The integration will use server-to-server communication over HTTPS.

Interface Requirement	Description
Communication Method	REST API
HTTP Method	POST
Protocol	HTTPS
Data Format	JSON
Integration Direction	BeeHive → Purple Bee
Authentication	Server-to-server authentication using environment-specific credentials
Purpose	Create a Purple Bee task from a confirmed BeeHive action item

Request Data

The API request must contain the information required to create and trace the Purple Bee task.

The request may include:

- BeeHive meeting UUID
- BeeHive action-item UUID
- Task title
- Task description
- Project UUID
- Assignee UUID
- Priority
- Due date
- Confirmation details
- Idempotency key

The API must only transfer information approved for the integration. Recordings, full transcripts, full AI summaries, private notes, secrets and unrelated participant information must not be included.

API Response

A successful response must provide:

- Transfer status
- Purple Bee task UUID
- Relevant response information

An unsuccessful response must provide an appropriate HTTP status and error information so that the cause of the failure can be identified.

Error Responses

The integration should handle common API errors, including:

- **400 Bad Request:** Invalid or incomplete request data.
- **401 Unauthorized:** Authentication failed or credentials are invalid.
- **403 Forbidden:** User or service does not have the required permission.
- **409 Conflict:** Duplicate or idempotency conflict.
- **5xx Server Error:** Temporary server-side or integration failure.

Performance Requirements

API Response Performance

The integration should provide a timely response when creating a Purple Bee task. The final response-time target must be confirmed during technical validation and testing.

Retry Performance

Temporary failures such as network or server errors may be retried. Permanent validation or authorisation errors must not be automatically retried.

Retry attempts must be limited to prevent repeated requests from creating unnecessary system load.

Duplicate Prevention

The integration must prevent duplicate task creation when the same action item is submitted more than once.

The idempotency key must follow the format:

beehive:{meeting_id}:{action_item_id}

A repeated request using the same key and request data must return the existing task rather than create a new task.

Availability

The integration must operate when the BeeHive, Purple Bee and required backend services are available. Integration failures must be recorded so that unsuccessful transfers can be investigated and reconciled.

Security Requirements

Authentication

All integration requests must be authenticated. The integration must use secure server-to-server authentication and environment-specific credentials.

Authentication credentials must not be exposed in source code or transferred through insecure channels.

Authorisation

Only the nominated Project Lead with an active Purple Bee Admin or Manager role may confirm an action item for transfer.

Purple Bee must re-check the confirmer's authority, membership, project and assignee information when processing the request.

Data in Transit

All communication between BeeHive, the integration service, and Purple Bee must use HTTPS/TLS to protect information while it is being transferred.

Data Protection and Privacy

Only information required for creating and tracing the Purple Bee task may cross the integration boundary.

The following information must not be transferred:

- Meeting recordings
- Full transcripts
- Full AI summaries
- Private or unshared notes
- Secrets or credentials
- Unrelated participant information

Secret Management

API keys, tokens, passwords, and other sensitive configuration values must be stored securely using appropriate environment-specific secret management.

Secrets must not be committed to source-control repositories.

Audit Logging

The integration must record sufficient information to support auditing and troubleshooting, including:

- Meeting UUID

- Action-item UUID
- Confirming user's identity
- Confirmation timestamp
- Transfer status
- Purple Bee task UUID
- Error information where applicable

Access Control

The integration must follow Purple Bee's role-based access control. Users must only be able to access and perform actions permitted by their assigned role.

Data Requirements

The integrated solution must maintain a clear separation of data ownership between BeeHive and Purple Bee. BeeHive remains the source of truth for meeting context, action item provenance, approved AI notes, confirmation information and transfer status. Purple Bee remains the source of truth for projects, profiles, created tasks, assignments notifications and task lifecycle.

Key data entities

Entity	Business Meaning	Source of truth
Meeting	Represents a Beehive meeting and its meeting context	BeeHive
Meeting Action Item	Represents an action identified from a meeting and approved for possible transfer	BeeHive
User/profile	Identifies users participating in or interacting with the system	Purple Bee
Project	Represents Purple Bee project receiving the task	Purple Bee
Team	Represents the Purple Bee team associated with a task	Purple Bee
Task	Represents the operational work created from a confirmed action item	Purple Bee
Integration receipt	Records the results / evidence of the integration transaction	Purple Bee
Notification	Represents the assignment notification associated with the created task	Purple Bee

Data Mapping

BeeHive to Purple Bee Data Mapping

BeeHive Data	Purple Bee Data	Purpose
meeting_id	Source meeting reference	Maintains traceability to the originating meeting
meeting_action_item_id	Action item/source reference	Identifies the original action item
Action item title	Task title	Provides task name
Action item description	Task description	Provides task details
Assignee UUID	Assignee/profile UUID	Identifies the Purple Bee user responsible for the task
Project UUID	Project UUID	Identifies the destination project
Priority	Task priority	Transfers the approved priority
Due Date	Task due date	Transfers the required due date
Tags	Task tags	Transfers approved task tags
Confirmer identity	Confirmation	Identifies who approved the transfer
Confirmation timestamp	Integration	Records when the action item was confirmed
Purple Bee task UUID	BeeHive transfer record	Maintains the link to the created task

Data Validation Requirements

The following validation requirements apply before and during task creation:

1. Every exportable action item must have a stable meeting_action_items_id UUID.
2. The title must be provided.
3. A project must be selected.
4. An assignee must be provided.
5. Priority must be provided.

6. The due date must not be in the past.
7. The confirmer must be the nominated project lead.
8. The confirmer must have an active Purple Bee Admin or Manager team role.
9. Purple Bee must re-check the confirmer's role and membership.
10. Purple Bee must validate the selected project and assignee
11. Purple Bee must derive team_id from project_id.
12. Conflicting client provided team information must be ignored or rejected.
13. Duplicate requests must be identified using the defined idempotency.
14. A changed request using an existing idempotency key must not create another task

Data Ownership and Access

Data Ownership

Data	Owner
Meeting context	BeeHive
Meeting action-item provenance	BeeHive
Confirmation record	BeeHive
Transfer status	BeeHive
Team information	Purple Bee
Project information	Purple Bee
User/profile information	Purple Bee
Created task	Purple Bee
Task lifecycle/status	Purple Bee
Assignment notification	Purple Bee
Integration receipt	Purple Bee

Access

Access to the integration data must follow the permissions of the respective systems. Only the nominated project lead with an active Purple Bee Admin or Manager team role may confirm an action item for transfer.

Recordings, full transcripts, full summaries, secrets and unrelated participant information must not be transferred through the integration.

Data Synchronisation and Conflict Handling

The first release uses one-way synchronisation from BeeHive to Purple Bee for task creation.

After a task has been created, Purple Bee owns the task lifecycle. Changes made to the Purple Bee task are not written back to BeeHive.

Duplicate task creation is prevented through the defined idempotency key:

`beehive:{meeting_id}:{action_item_id}`

If the same key and request hash are received again, the existing task is returned. If the same key is received with a different request, the request returns a 409 response.

Purple Bee derives team_id from the selected project rather than accepting a conflicting team value from the client.

Reconciliation must not automatically create a replacement task when an existing task is missing or deliberately deleted.

Audit and History Requirements

The integrated solution must retain sufficient information to establish traceability between the original BeeHive action_item and resulting Purple Bee task.

The integrated solution must retain sufficient information to establish traceability between the original BeeHive action item and the resulting Purple Bee task.

The traceability information includes:

- Source meeting UUID
- BeeHive action-item UUID
- Confirmer identity
- Confirmation timestamp
- Transfer status
- Purple Bee task UUID
- Integration receipt
- Relevant failure or retry information

The integration must allow the business and technical teams to determine whether a confirmed action item was successfully transferred and identify unresolved failures. This supports the documented requirement for traceability and operational reconciliation.

Operational Requirements

Integration Processing

Confirmed action items must be queued for delivery and processed by the integration service.

Failure Handling

Permanent validation errors must return the item for correction.

Transient errors may be retried.

The documented retry schedule is:

- 1 minute
- 5 minutes
- 15 minutes
- 1 hour
- 6 hours

Unresolved records must be available for investigation rather than being silently replaced.

Monitoring

The integration must provide sufficient operational information to identify:

- Successful transfers
- Failed transfers
- Retry attempts
- Unresolved records
- Duplicate requests
- Integration receipts

Reconciliation

The integration must support reconciliation between BeeHive transfer records and Purple Bee task/integration records.

Database and Hosting

The two systems remain on separate Supabase-backed environments. Direct database writes and cross-database foreign keys are not part of the integration design.

Integration & Dependency Requirements

Integration Matrix

Source	Target	Data	Trigger	Frequency	Failure Handling
BeeHive	Purple Bee Integration API	Confirmed action-item information, task details and relevant UUIDs	Project Lead confirms an action item	On demand	Validation errors are returned for correction; approved temporary failures are retried

Integration Direction

Phase 1 supports a one-way integration from BeeHive to Purple Bee.

The integration flow is:

BeeHive → Integration API/Service → Purple Bee

Purple Bee task updates are not synchronised back to BeeHive during Phase 1.

Data Mapping

The confirmed BeeHive action item is mapped to a Purple Bee task.

BeeHive Information	Purple Bee Information
Action Item UUID	Source Action Item UUID
Meeting UUID	Source Meeting UUID
Action Item Title	Task Title
Action Item Description	Task Description
Project UUID	Project
Assignee UUID	Assignee
Priority	Task Priority
Due Date	Task Due Date

Synchronisation

The integration uses request-based transfer rather than continuous two-way synchronisation.

Once an action item has been confirmed, the integration sends the approved information to Purple Bee. Purple Bee becomes the system of record for the created task and its future task lifecycle**.**

Duplicate Prevention

Each transferred action item must use a unique idempotency key:

beehive:{meeting_id}:{action_item_id}

This prevents repeated requests from creating duplicate Purple Bee tasks.

Failure and Retry Handling

The integration must distinguish between permanent and temporary failures.

- Permanent validation errors must be returned for correction.
- Authentication and authorisation failures must be rejected.
- Temporary server or network failures may be retried.
- Duplicate requests must return the existing task or an appropriate conflict response.
- Failed transfers must be logged for investigation.

Dependencies

The integration depends on:

- BeeHive backend services
- Purple Bee backend/API
- Supabase services
- Authentication services
- HTTPS/network connectivity
- Environment-specific configuration and secrets
- Required database services

BeeHive & Purple Bee Integration

Integration matrix

Integration	Source	Target	Data	Trigger	Frequency	Failure handling
INT-001	Beehive	Purple Bee	Permitted team, project and member UUID context	Review setup or mapping revalidation	On-demand HTTP GET; cache policy validated in staging	Return unavailable state; never guess identity mapping.
INT-002	Beehive	Purple Bee	Confirmed task fields and minimum meeting/action provenance	Nominated Admin or Manager confirmation plus outbox claim	Event-driven asynchronous HTTP POST	Idempotent retry, correction, reconciliation and dead-letter handling.
INT-003	Purple Bee	Beehive	Created/replayed task UUID, receipt UUID and result code	HTTP response	Per request	Safe resend with same key if response is lost.
INT-004	Purple Bee database	Purple Bee user	Task-assigned notification	Atomic task transaction commit	Existing real-time notification path	Atomic with task or transaction fails.
INT-005	Both systems	Operations	Source status, receipt/task UUID and mismatch result	Daily schedule or authorised manual run	Scheduled and on demand	Record incomplete run; never recreate silently.

Questions the SRS must answer

Required question	Completed answer
Which system owns each important piece of data?	Beehive owns meeting data, canonical action, human confirmation and transfer state. Purple Bee owns teams, projects, profiles, receipts, tasks, notifications and future task state.
What causes data to move?	Only explicit confirmation of one action by the nominated project lead with active Admin or Manager team authority creates an outbox event. Meeting/AI completion only prepares candidates.
What is the synchronisation method?	Event-driven asynchronous one-way creation through versioned HTTP APIs over HTTPS. Context lookup is on demand. Reconciliation is daily and on demand.
What happens if a system is unavailable?	Confirmation remains stored in the Beehive outbox. The worker retries approved transient failures. The UI shows queued/retryable state. No task is claimed until Purple Bee returns creation or replay success.
How are duplicates prevented?	A stable meeting/action idempotency key, request hash, unique Purple Bee receipt and atomic task transaction guarantee at-most-one target task for one create intent.
How are conflicting updates resolved?	Optimistic action versioning prevents concurrent confirmation. Same key with changed body returns 409. Purple Bee remains authoritative for target relationships and later task changes.
How is authentication handled?	Beehive users use normal Supabase sessions. Server-to-server HTTP calls use HTTPS plus environment-specific HMAC key ID, timestamp, nonce, and signature. Purple Bee re-checks human role and membership.
How are failures retried and monitored?	Timeout, 429, and 5xx receive bounded exponential backoff and jitter. Permanent 4xx errors stop. Metrics, correlation UUIDs, sanitised errors, alerts and a dead-letter state support operations.
How does an administrator know integration failed?	Beehive shows item transfer status. Operations receives alerts for old queued events, authentication failures, conflicts, repeated failures and reconciliation mismatches.

How is data reconciled?	A scheduled job compares Beehive successful actions with Purple Bee receipts/tasks by immutable source UUID and idempotency key, records mismatches, and requires human decisions for missing tasks.

External dependencies

Dependency	System	Purpose	Owner	Authentication	Limitations or cost	Fallback
Supabase projects	Both	Authentication, PostgreSQL, RLS, storage where used, and Purple Bee Edge Functions.	Application owners	User JWTs and HMAC service authentication	Regions, limits, backup and SLA require staging validation.	No production release without verified staging projects.
LiveKit	Beehive	Meeting rooms and verified lifecycle events.	Beehive owner	Webhook verification	Availability and event retry policy require confirmation.	Sweep meeting state and use existing recovery.
Recording transcription and AI providers	Beehive	Optional recording, transcription, summary and suggested actions.	Beehive owner	Provider server credentials	Provider, quota, cost and SLA require confirmation.	Agenda-only review remains available.
Render hosting environments	Both	Application and worker execution.	DevOps	Environment-managed secrets	Network, scaling and availability require confirmation.	Feature flag and controlled rollback.
Notification or email service	Purple Bee	Existing task assignment communication where configured.	Purple Bee owner	Existing service credentials	Delivery limits and costs depend on deployed provider.	Task and receipt remain traceable if external delivery fails.

Requirement IDs, Priority & Traceability

Traceability and Acceptance

For example:

Requirement ID	Requirement	Source	Priority	Acceptance Criteria	Related Use Case
FR-001	Confirmed action items can be transferred to Purple Bee as tasks.	Business Requirement	High	A confirmed action item creates a corresponding Purple Bee task.	UC-01
FR-002	Only authorised Project Leads can confirm action items.	BR-003	High	Unauthorised users are prevented from confirming.	UC-01
FR-003	API communication must use HTTPS.	BR-018	High	All integration requests use HTTPS/TLS.	UC-01
FR-004	Required task information	BR-008	High	Missing or invalid information prevents task creation.	UC-02

	must be validated.				
FR-005	Duplicate tasks must be prevented.	BR-010, BR-011	High	Repeated requests do not create duplicate tasks.	UC-01
FR-006	The Purple Bee team must be derived from the selected project.	BR-009	High	The created task uses the project's associated team.	UC-03
FR-007	Successful task creation must return the Purple Bee task UUID.	Integration Requirement	High	The task UUID is returned and recorded.	UC-03
FR-008	Approved temporary failures must be retried.	BR-016	High	Temporary failures are retried according to the retry policy.	UC-05
FR-009	Prohibited or private information must not be transferred.	BR-014	High	Recordings, transcripts, private notes and secrets are excluded.	UC-01
FR-010	Integration activity must be auditable.	Security Requirement	High	Confirmation, transfer status and task UUID are recorded.	UC-04

Acceptance Criteria

Then underneath the matrix, have the overall acceptance criteria, such as:

- Confirmed action items are successfully transferred to Purple Bee.
- Only authorised Project Leads can confirm action items.

- Required information is validated before task creation.
- Duplicate task creation is prevented.
- The Purple Bee task UUID is returned and recorded.
- Temporary failures are handled according to the retry policy.
- Transfer activity is recorded for traceability and investigation.
- Private and prohibited information is not transferred.
- API communication uses secure HTTPS and authentication.
- Purple Bee remains responsible for the task lifecycle.

Open Questions, Risks & Deliverables

Open Questions

ID	Question/Decision	Impact	Status
OQ-001	What are the final required Purple Bee task fields and their exact validation constraints?	High	Open
OQ-002	Which Purple Bee role field is authoritative for confirming Admin/Manager authority?	High	Open
OQ-003	What are the final BeeHive-to-Purple-Bee API endpoint URLs and API contract details?	High	Open
OQ-004	What are the required retention periods for integration and audit records?	Medium	Open
OQ-005	What are the expected integration volumes and API rate limits?	Medium	Open
OQ-006	What performance and availability targets should the integration meet?	Medium	Open
OQ-007	Which Purple Bee project statuses allow new tasks to be created?	Medium	Open
OQ-008	What is the final behaviour for Start Now meetings where AI recording/processing is not used?	Medium	Open

Risks and Constraints

ID	Risk / Constraint	Impact	Mitigation
R-001	Incorrect or incomplete action-item data could result in failed task creation.	High	Validate required fields before confirmation and again in Purple Bee.
R-002	Duplicate requests could create duplicate tasks.	High	Use the defined idempotency key and request-hash validation.
R-003	One system may become temporarily unavailable during transfer.	High	Retry approved transient failures and retain transfer status.
R-004	Incorrect role or membership information could allow unauthorised confirmation.	High	Purple Bee independently validates the confirmer's role and membership.
R-005	Sensitive meeting information could be transferred unintentionally.	High	Restrict the API payload to approved task-related information.
R-006	Changes to Purple Bee task data after creation could cause inconsistency if	Medium	First release is explicitly one-way

	synchronisation is expected.		and Purple Bee owns the task lifecycle.
R-007	Missing or deleted Purple Bee tasks could lead to incorrect automatic recreation.	Medium	Reconciliation must not silently create replacement tasks.
R-008	Unknown API limits or performance requirements could affect implementation.	Medium	Confirm volumes, rate limits and performance targets before final implementation.

Constraints

- The integration is planned but not yet implemented or tested; the SRS defines the approved implementation contract.
- HTTP APIs over HTTPS are confirmed. The endpoints and backing database structures must still be built and verified in staging.
- Beehive agenda/AI sources do not currently provide one stable confirmed action entity, so a canonical action-item structure is required.
- Purple Bee currently has no protected external task-creation API in the inspected repository.
- Production data and secrets may not be used in development or copied into documentation.
- The first phase is one-way task creation only.